



EC 503
MACROECONOMICS I
FALL 2018
SYLLABUS

Course Description: EC 503 Macroeconomics I is the first course of the graduate-level macroeconomics sequence of the M.A. program in Economics at Boğaziçi University. The course is divided into two parts: the first part, taught by [Orhan Torul](#) will focus mainly on the modern macroeconomic methodology at graduate-level macroeconomics throughout a discussion over consumption, and the second part, taught by [Murat Koyuncu](#) will concentrate on continuous time models and growth theory. Below please find the details of the two parts.

PART I

Instructor: [Orhan Torul](#)

Time and Location: M12 9:00-10:50, TBA & W12 9:00-10:50, TBA

Telephone: 212-359-6677

Email: orhan.torul@boun.edu.tr

Office Hours: M78 15:00-17:00 or by appointment, NB222A

Course Website: [Boğaziçi University Department of Economics](#) → [Courses](#) → [EC 503.01](#)

Teaching Assistant: [TBA](#)

Course Objectives

This is the first part of the second graduate-level course of the macroeconomics sequence in the M.A. program. This part intends to accomplish two major goals: the first objective is to familiarize students with advanced economic tools, such as dynamic programming under certainty and uncertainty, which we will cover throughout our discussion over theoretical and empirical foundations of consumption. The second objective of this part is to study a number of contemporary economic questions which cannot be addressed using the representative-agent paradigm. For this purpose, we will study heterogeneous-agent models to discuss several issues, such as the determination of prices in the presence of incomplete markets and politico-economic equilibrium over policies, such as taxes.

Requirements

This is a graduate level course, thus graduate standing is a prerequisite for the course. Exceptions are to be discussed by the instructors of the course.

Textbook

“Recursive Macroeconomic Theory”, 3rd Edition by Lars Ljungqvist and Thomas J. Sargent, MIT Press, August 2012. (L+S hereafter)

Grading Policy

This part constitutes to half of your grade in EC503. Your overall grade from this part will be calculated based on your grades from problem sets, unannounced pop quizzes and class participation and the midterm exam, along with your grade from the summer preparation course on macroeconomics. The weights are as follows:

- **Problem Sets:** 5%
- **Pop Quizzes + Class Participation:** 10%
- **Midterm Exam:** 35%

As a reminder, the standard catalog grading protocol accepted at Boğaziçi University is as follows:

Grade Scale			
Letter Grade	Interpretation	Weight	Grade Percentage
AA	Excellent	4.0	90%
BA	Good-Excellent	3.5	85-89%
BB	Good	3.0	80-84%
CB	Passing-Good	2.5	75-79%
CC	Passing	2.0	70-74%
F	Failed	0.0	0-59%

Yet, we reserve the right to modify criteria for letter grades based on distribution of cumulative grades of the class.

Please check the withdrawal dates on the [Academic Calendar](#)

- **Midterm Exam (*Tentative Date: TBA*):** The midterm exam will cover all subjects up to the exam date. Midterm exam will be given either during the class time or at 17:00; either way it is imperative for you to be present in class on time, since extra time will not be given for late arrivals.
- **Attendance:** Regular attendance is **required**. Students who do not attend classes regularly may not only lose all the credit from unannounced quizzes, but also can get lower grades due to **poor class participation**.

- **Problem Sets:** There will be a number of problem sets (5 ± 1), all of which will be graded, each with equal weight. Problem sets will be posted online, and their due dates will be announced on the course web site along with problem sets. Please check course web site frequently so as not to miss deadlines.
- **Make-up Exam Policy:** All students are required to attend both the midterm exam. Make-up midterm exam will be given only to those students whose excuses comply with the University policy. If you believe you are eligible to take a make-up exam, please inform me and provide the appropriate documentation as soon as possible, no later than one week (7 calendar days) after the exam. Make-up exam requests should be directed to me in person with proper documentation (together with a clear written explanation of the reason of absence and valid excuse documents). As for your information, historically make-up exam grades have been *considerably* lower than regular exams.
- **Exam Re-grade Policy:** If, after going over your exam and the exam solutions, you believe some of your solutions were more correct than originally judged to be, you may submit, in writing, an exam re-grade request. Your re-grade request must specify which solutions(s) you believe were not graded appropriately and a substantive explanation for why you believe your solutions is more correct than originally judged. Your entire exam is subject to re-grade, thereby raising the possibility of a lower overall grade. Exam re-grade requests should be submitted no later than one week (7 calendar days) after exams are announced, and this deadline applies even if you are not present in class the day graded exams are announced (it is your responsibility to be responsive to your graded exams in a timely manner).

Problem Sessions

There will be weekly/biweekly problem sessions held by your teaching assistants (dates and locations to be announced). Attendance is very strongly recommended although not required.

Communication

E-mail will be used as the primary means of communication outside the classroom, and I will be sending e-mails whenever necessary to inform you on updates regarding the course. Please make sure that the e-mail address you have at the Registrar's Office is up-to-date so that you will not miss any of the announcements.

Accommodations

Students who require special accommodations for exams must get in touch with me within the first two weeks of class.

Academic Integrity

The graduate program of the Department of Economics is conducted within the framework of the Student Discipline Regulations by the Turkish Council of Higher Education

(TCHE, Yükseköğretim Kurulu, YÖK in Turkish), and rules accepted by the Boğaziçi University Committee on Ethical Conduct.

For student discipline regulations by the TCHE (YÖK), see (in Turkish): bit.ly/yokdisiplin

For the ethics code accepted by Boğaziçi University, see: bit.ly/bounethics

EC 103 Orientation to Economics course also discusses basics of integrity. For a brief refresher, see bit.ly/ec103notes

Copyrights

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Tentative Course Outline

Week 1:	Life Cycle/Permanent Income Hypothesis under Certainty
Week 2:	Decision Making Under Uncertainty/ The LCPIH Under Uncertainty
Week 3:	Decision Making Under Uncertainty/ The LCPIH Under Uncertainty (cont'd)
Week 4:	Review and Testing of the Complete Markets Economy
Week 5:	Certainty Equivalence, Non-Negativity Constraints and Precautionary Savings
Week 6:	Macroeconomic Models with Incomplete Markets and Heterogeneity
Week 7:	Political-Economy in Macroeconomics

References

The Life Cycle/Permanent Income Hypothesis and Variants: Theory

- L+S: Chapter 1; Chapter 2 (you can skim Sections 2.5, 2.7, 2.8); Chapter 3; Chapter 4; Chapter 5 (you can skim Sections 5.4 & 5.6); Chapter 8 (you can skim Sections 8.7–8.12); Chapter 17; Chapter 18 (you can skim sections 18.8–18.12)
- Deaton, Angus (1991), “Saving and Liquidity Constraints,” *Econometrica*, 59, 1221–1248.
- Zeldes, Steven (1989), “Optimal Consumption with Stochastic Income: Deviations from Certainty Equivalence,” *Quarterly Journal of Economics*, 104, 275–298.

Consumption: Empirical Evidence

- Agarwal, Sumit; Liu, Chunlin; and Nicholas Souleles (2007), “The Reaction of Consumer Spending and Debt to Tax Rebates? Evidence from Consumer Credit Data,” *Journal of Political Economy*, 115(6), 986–1019.
- Aguiar, Mark and Eric Hurst (2005), “Consumption Versus Expenditure,” *Journal of Political Economy*, 113 (5), 919–948.

- Aguiar, Mark and Eric Hurst (2007), "Life-Cycle Prices and Production," *American Economic Review*, 97(5), December, 1533-1559.
- Card, David; Chetty, Raj and Andrea Weber (2007), "Cash-on-Hand and Competing Models of Intertemporal Behavior: New Evidence from the Labor Market," *Quarterly Journal of Economics*, November, 1511-1560.
- Cochrane, John (1991), "A Simple Test of Consumption Insurance," *Journal of Political Economy*, 99 (5), October, 957-975.
- Hall, Robert (1978), "Stochastic Implications of the Life Cycle-Permanent Income Hypothesis: Theory and Evidence," *Journal of Political Economy*, 86, 971-987.
- Hsieh, Chang-Tai (2003), "Do Consumers React to Anticipated Income Changes? Evidence from the Alaska Permanent Fund," *American Economic Review*, 93 (1), 397-405.
- Scholz, John Karl; Seshadri, Ananth; and Surachai Khitatrakun (2006), "Are Americans Saving 'Optimally' for Retirement?" *Journal of Political Economy*, 114, 607-43.
- Shea, John (1995), "Union Contracts and the Life Cycle-Permanent Income Hypothesis," *American Economic Review*, 86, 186-200.
- Stephens, Melvin (2003). "3rd of the Month': Do Social Security Recipients Smooth Consumption Between Checks?" *American Economic Review*, 93(1), 406-422.
- Wilcox, David (1989), "Social Security Benefits, Consumption Expenditure and the Life-Cycle Hypothesis," *Journal of Political Economy*, 97, 288-303.
- Zeldes, Stephen P. (1989), "Consumption and Liquidity Constraints: An Empirical Investigation," *Journal of Political Economy*, 97, 305-346

Asset Pricing: A Basic Introduction

- Barro, Robert (2006), "Rare Disasters and Asset Markets in the Twentieth Century," *Quarterly Journal of Economics*, 121, 823-866.
- Brunnermeier, Markus and Stefan Nagel (2008), "Do Wealth Fluctuations Generate Time-Varying Risk Aversion? Micro-Evidence on Individuals' Asset Allocation," *American Economic Review*, 98 (3), 713-736.
- Kocherlakota, Narayana (1996), "The Equity Premium: It's Still a Puzzle," *Journal of Economic Literature*, 34, 42-71.

Macroeconomic Models with Incomplete Markets

- Aiyagari, S. Rao, (1994), "Uninsured Idiosyncratic Risk and Aggregate Saving," *Quarterly Journal of Economics*, 109, 659-684.
- Heathcote, Johnathan; Storesletten, Kjetil and Gianluca Violante (2009), "Quantitative Macroeconomics with Heterogeneous Households," *Annual Review of Economics*, 1, 319-354.

- Huggett, Mark (1993), “The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies,” *Journal of Economic Dynamics and Control*, 17, 953-969.
- Krusell, Per and Anthony A. Smith Jr., (1998), “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, 106(5), 867-896.
- Rios-Rull, Jose-Victor (1995), “Models with heterogeneous agents,” *Frontiers of Business Cycle Research*, Princeton University Press, 98-125.

Political Economy of Economic Growth

- Meltzer, Allan H. and Scott F. Richard (1981), “A Rational Theory of the Size of Government,” *Journal of Political Economy*, 89, 914-27.
- Persson, Torsten and Guido Tabellini (1994), “Is Inequality Harmful for Growth?,” *American Economic Review*, 84, 600-621.

Extra: Cross Sectional Facts and Transatlantic Differences

- Alesina, Alberto, Edward Glaeser (2004), “Fighting Poverty in the U.S. and Europe A: World of Difference”, Oxford, UK: Oxford University Press.
- Alesina, Alberto, Edward Glaeser and Bruce Sacerdote (2005), “Work and Leisure in the United States and Europe: Why So Different?,” *NBER Macroeconomics Annual*, 20, 1-64.
- Alesina, Alberto, and George-Marios Angeletos (2005), “Fairness and Redistribution,” *American Economic Review*, 95, 960-980.
- Bénabou, Roland and Jean Tirole (2006), “Belief in a Just World and Redistributive Politics,” *The Quarterly Journal of Economics*, 121, 699-746.
- Blundell, Richard and Ben Etheridge (2010), “Consumption, Income and Earnings Inequality in Britain,” *Review of Economic Dynamics*, 13(1), pages 76-102.
- Domeij, David and Martin Floden (2010), “Inequality Trends in Sweden 1978-2004,” *Review of Economic Dynamics*, 13(1), 179-208.
- Fuchs-Schuendeln, Nicola; Krueger, Dirk and Mathias Sommer (2010), “Inequality Trends for Germany in the Last Two Decades: A Tale of Two Countries” *Review of Economic Dynamics*, 13(1), 03-132.
- Heathcote, Jonathan; Perri, Fabrizio and Giovanni L. Violante (2010), “Unequal We Stand: An Empirical Analysis of Economic Inequality in the United States: 1967-2006,” *Review of Economic Dynamics*, 13(1), 15-51.
- Jappelli, Tullio and Luigi Pistaferri (2010), “Does Consumption Inequality Track Income Inequality in Italy?” *Review of Economic Dynamics*, 13(1), 133-153.
- Pijoan-Mas Josep and Virginia Sanchez-Marcos (2010), “Spain is Different: Falling Trends of Inequality,” *Review of Economic Dynamics*, 13(1), 154-178.
- Piketty, Thomas and Gabriel Zucman (2013), “Capital is Back: Wealth-Income Ratios in Rich Countries, 1700-2010,” CEPR Discussion Papers 9588.

PART II

Instructor: [Murat Koyuncu](#)

Time and Location: M12 9:00-10:50, TBA & W12 9:00-10:50, TBA

Telephone: 212-359-7640

Email: [Murat Koyuncu](#)

Office Hours: TBA or by appointment, NB224

Course Description

This part intends to accomplish two major goals: the first objective is to familiarize students with mathematical methods of continuous time analysis. The second objective is to familiarize the students with growth theory.

Prerequisites

Even though we will discuss the basic material in class, it is absolutely necessary that you are familiar with basic notions of calculus, constrained optimization, probability and statistics. This is a graduate course in economics.

Textbook

Lecture notes are the primary source for this course. There is no required text. However, for some of the topics, the following books might be useful:

- Acemoglu, Daron, 2009, Introduction to Modern Economic Growth, Princeton Univ. Press.
- Barro, Robert and Xavier, Sala-i Martin, 2004, Economic Growth, MIT Press.

There might be additional required readings as we go along, those readings will be posted on the course website.

Course requirements

Class participation

Regular attendance is expected.

Short Quizzes

There will be several in-class quizzes during the semester. Some of them might be announced before hand, but **most of them will be unannounced**.

Assignments

There will be several homework assignments. Due dates of each assignment will be announced ahead of time during the semester. NO assignment is accepted after the due date.

Exams

Note that there will be no early exams and no re-writes. Final exam will cover only the topics in this part.

Assessment and grading scale

The weights for your score in this part will be as follows:

Concept	Percentage of your grade
Assignments and In-class Quizzes:	15%
Final Exam	35%

Standard grading scale is given above in the first part of the syllabus. I reserve the right to scale it up or down depending on the average and standard deviation of the exam grades.

Tentative Course Outline

Week 1:	Dynamic Optimization Methods in Continuous Time
Week 2:	Application: Ramsey Model in Continuous Time
Week 3:	Stylized Facts of Growth & Solow Model
Week 4:	Exogenous Growth (Ramsey-Cass-Koopmans) Model
Week 5:	Exogenous Growth (ctd.)
Week 6:	Heterogeneous Agent Models with Complete Markets

Reading List for Heterogeneous Agent Models with Complete Asset Markets

- Caselli, F., and Ventura, J. (2000). “A representative consumer theory of distribution,” *American Economic Review*, Vol. 90, pp. 909–926.
- Chatterjee, S. (1994). “Transitional dynamics and the distribution of wealth in a neoclassical growth model”, *Journal of Public Economics*, Volume 54, Issue 1, pp. 97–119.
- García-Peñalosa, C. and Turnovsky, S. J. (2006). “Growth and Income Inequality: A Canonical Model” *Economic Theory*, Vol. 28, No. 1, pp. 25–49.
- Ogaki, M. (2003). “Aggregation under Complete Markets”, *Review of Economic Dynamics*, Volume 6, Issue 4, pp. 977–986.